

Task 2: Baseline Models & Proposed GNN

1. Experimental Setup

- Due to the **large dataset size (683,547 samples)** and long training time, **50% of the dataset (341,773 samples)** was used for all experiments.
- Applied a **subject-independent train/validation/test split (80/10/10)**.
- Performed **label encoding, feature scaling, and preprocessing** without data leakage.

2. Baseline Evaluation

- Evaluated **Logistic Regression, k-NN, Decision Tree, Random Forest, XGBoost, LightGBM, MLP, and 1D-CNN**.
- **Random Forest** was the **best baseline**:

Accuracy: 84.28%

Macro F1-score: 0.8067

- Models were compared using **Accuracy, Precision, Recall, F1-score, and Training Time**.

3. Proposed GNN

- Built a **Graph Neural Network (GNN)** where the **12 sEMG channels** were treated as graph nodes with fully connected edges to learn spatial relationships.
- **Architecture:** Input → Graph → GCN Layers → Global Pooling → MLP Classifier → Output.
- Implemented using **PyTorch Geometric** and trained for **15 epochs**.

4. Initial Results

- **GNN Performance:**

Accuracy: 69.13%

Macro F1-score: 0.5323

- The initial GNN outperformed simple models but **did not surpass the Random Forest baseline**.
- **Future Work:** Improve graph construction using dynamic edge weights, Graph Attention Networks (GAT), and GPU-based training to improve performance.